Lecture 3: Ordinary Least Squares

In this lecture, we will discuss **ordinary least squares (OLS)**. This is the main estimator that we will discuss in this course, and understanding OLS will be lay the foundation for understanding other estimators.

Motivation: Loss Functions

In the past, you may have heard the phrase "best-fit line" to refer to a line that "best fits" the points on a scatter plot. Although we will justify OLS as a best-fit line, we first need to determine a notion of fit: what determines how well a line "fits" the data?

In general, we can evaluate a model's fit by thinking about a **loss function**, $L\left(Y_i, \hat{Y}_i\right)$, where Y_i is the actual value of the outcome and $\hat{Y}_i$ is the fitted value (the model's prediction of the outcome). The "best-fit" model minimizes the expected loss, under some loss function.

What loss function should we pick? Different loss functions will yield different estimators, which have different properties. To see this, imagine that we had a very simple model with no regressors, so that $\hat{Y}_i$ is just a number, which we will call θ. If the loss function is $L\left(Y_i, \hat{Y}_i\right) = \left(Y_i - \hat{Y}_i\right)^2$, then our best-fit value of θ will be:

$$
\begin{aligned}
\theta^{LS} &= \underset{\theta}{\operatorname{argmin}} \frac{1}{N}\sum_{i=1}^{N}\left(Y_i - \theta^{LS}\right)^2 \\
\implies 0 &= \frac{1}{N}\sum_{i=1}^{N} 2\cdot\left(Y_i - \theta^{LS}\right) \\
\implies \theta^{LS} &= \frac{1}{N}\sum_{i=1}^{N} Y_i = \bar{Y}
\end{aligned}
$$

Thus, under a quadratic loss function, the best-fit ("least-squares") value of θ^{LS} is the mean. Other loss functions will lead to other estimators. If the loss function were $L\left(Y_i, \hat{Y}_i\right) = \left(\log Y_i - \log \hat{Y}_i\right)^2$, then the best-fit value would be the geometric mean. If the loss function were $L\left(Y_i, \hat{Y}_i\right) = \left|Y_i - \hat{Y}_i\right|$, then the best-fit value would the median. Other loss functions, such as $L\left(Y_i, \hat{Y}_i\right) = \left(Y_i - \hat{Y}_i\right)^4$, tend to lead to less intuitive estimators.

Why then do we focus on the quadratic loss function, and on minimizing the sum of squares? We do this because least squares gives us estimators that depend on means. In our simple model without regressors, least-squares minimization gives us the mean of the outcome. In a model with regressors, the formula is slightly more complicated, but it will still be a function of the means of some objects in the data. This is convenient because means are well-behaved. The sample mean is an unbiased estimator of the true mean, and, as we saw in our lecture on asymptotics, the sample mean is a consistent and asymptotically normal estimator of the true mean. We will use these facts to prove that estimators based on least squares are also well-behaved.

Next, we will work out the ordinary least squares model in models with regressors: we first study the case of a single regressor, and then the general case with multiple regressors.

Least Squares with a Single Regressor

We will begin with the case of a single regressor. This model is sometimes called simple linear regression, univariate regression, or bivariate regression (these last two can be very confusing!). We write this model as follows:

$$Y_i = \alpha + \beta \cdot X_i + \varepsilon_i$$

where Y_i is the outcome variable, X_i is the regressor, the coefficient α is an intercept, the coefficient β is a slope, and ε_i is an error term. The outcome variable, Y_i, is also sometimes known as the dependent variable

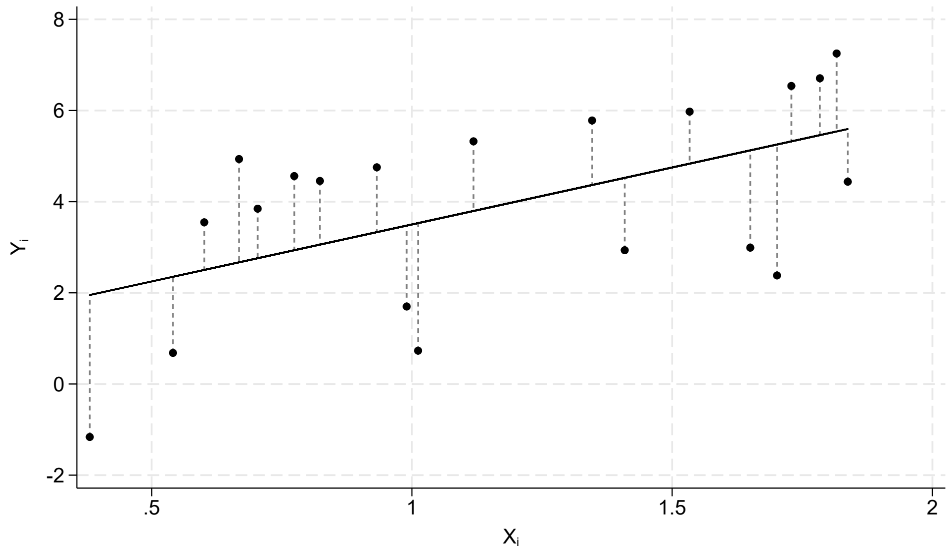

Figure 1: Visual Representation of Ordinary Least Squares

or left-hand-side variable, while the regressor, X_i, is also sometimes known as the independent variable or right-hand-side variable.

The least-squares estimator solves the following minimization problem:

$$\min_{\alpha,\beta} \frac{1}{N} \sum_{i=1}^{N} (Y_i - \alpha - \beta \cdot X_i)^2$$

that is, the least-squares estimator selects an intercept, α, and a slope, β, to minimize the sum of squared errors. For additional clarity, we can refer to Figure 1: the solid line depicts the least-squares estimates $\hat{\alpha}$, $\hat{\beta}$, while the deviations from that line are depicted with dashed vertical lines.

To find the α and β that minimizes the sum of squared errors, we can take first-order conditions. This yields:

$$0 = \frac{1}{N} \sum_{i=1}^{N} \left(Y_i - \hat{\alpha} - \hat{\beta} \cdot X_i\right) \cdot 2 \cdot (-1)$$
$$0 = \frac{1}{N} \sum_{i=1}^{N} \left(Y_i - \hat{\alpha} - \hat{\beta} \cdot X_i\right) \cdot 2 \cdot (-X_i)$$

Which simplifies to:

$$\hat{\alpha} = \bar{Y} - \hat{\beta} \cdot \bar{X}$$
$$0 = \frac{1}{N} \sum_{i=1}^{N} \left(X_i \left(Y_i - \hat{\alpha} - \hat{\beta} \cdot X_i\right)\right)$$

Note that the condition for $\hat{\alpha}$ implies that the regression line must pass through $(\bar{X}, \bar{Y})$. Plugging in our formula for $\hat{\alpha}$ then lets us solve for $\hat{\beta}$:

$$
\begin{aligned}
0 &= \frac{1}{N}\sum_{i=1}^{N}\left(X_i\left(Y_i - \bar{Y} + \hat{\beta}\cdot\bar{X} - \hat{\beta}\cdot X_i\right)\right) \\
\implies \hat{\beta} &= \frac{\frac{1}{N}\sum_{i=1}^{N} X_i\left(Y_i - \bar{Y}\right)}{\frac{1}{N}\sum_{i=1}^{N} X_i\left(X_i - \bar{X}\right)} \\
&= \frac{\widehat{\text{Cov}}\left(X_i, Y_i\right)}{\widehat{\text{Var}}\left(X_i\right)}
\end{aligned}
$$

It turns out that what looked complicated (the coefficients that minimize the sum of squared errors) can be computed as a very simple function of the data!

> **Exercise 1.** Let c be a constant. Show that $\mathbb{E}\left[c\cdot\left(Y_i - \mathbb{E}\left[Y_i\right]\right)\right] = 0$.
>
> **Exercise 2.** Show that $\mathbb{E}\left[X_i\left(X_i - \mathbb{E}\left[X_i\right]\right)\right] = \text{Var}\left(X_i\right)$, and that $\mathbb{E}\left[X_i\left(Y_i - \mathbb{E}\left[Y_i\right]\right)\right] = \text{Cov}\left(X_i\right)$. (The result from the previous exercise is useful for this).

To further our understanding of what least squares is doing, let's return to our first-order conditions. We will use the notation $\hat{\varepsilon}_i := \left(Y_i - \hat{\alpha} - \hat{\beta}\cdot X_i\right)$ to denote the **estimated residuals**.[1] We can then rewrite and simplify the first-order conditions as

$$
\begin{aligned}
0 &= \frac{1}{N}\sum_{i=1}^{N}\hat{\varepsilon}_i \\
0 &= \frac{1}{N}\sum_{i=1}^{N}\hat{\varepsilon}_i\cdot X_i
\end{aligned}
$$

The first states that OLS sets the mean of the estimated residual to zero, and the second states that OLS sets the mean of $\hat{\varepsilon}_i \cdot X_i$ to zero as well. Since OLS sets $\hat{\varepsilon}_i$ to have mean zero, we can also rewrite these equations in terms of covariance:

$$
\begin{aligned}
0 &= \frac{1}{N}\sum_{i=1}^{N}\hat{\varepsilon}_i \\
0 &= \widehat{\text{Cov}}\left(\hat{\varepsilon}_i, X_i\right)
\end{aligned}
$$

In this form, the equations state that $\hat{\varepsilon}_i$ must have zero mean and also zero covariance with the regressor. In either form, these equations are often called **orthogonality conditions**.

> **Exercise 3.** Suppose that we run a regression of Y_i on X_i (a regression with Y_i as the outcome and X_i as the regressor), and call our least-squares estimate of the slope $\hat{\beta}_{Y,X}$. We then run a regression of X_i on Y_i (now X_i is the outcome and Y_i is the regressor), and call our least-squares estimate of the slope $\hat{\beta}_{X,Y}$. Will these be the same? Solve for the ratio of $\hat{\beta}_{Y,X}/\hat{\beta}_{X,Y}$.
>
> **Exercise 4.** Now suppose that, in our sample, $\text{Var}\left(X_i\right) = \text{Var}\left(Y_i\right)$. Show that both $\hat{\beta}_{Y,X}$ and $\hat{\beta}_{X,Y}$ are equal to the correlation coefficient.

Why are these orthogonality conditions sufficient to pin down the unique least-squares line? The answer is that we have two equations and two unknown parameters. To see this visually, in Figure 2, we can see that there are many lines that satisfy the condition that the the estimated residuals have zero mean, $\frac{1}{N}\sum_{i=1}^{N}\hat{\varepsilon}_i = 0$. This is equivalent to requiring that the line must pass through $\left(\bar{X}, \bar{Y}\right)$. There are an infinite

[1] Many people draw a distinction between the "error term" ε_i , and the "residuals" $\hat{\varepsilon}_i$. However, others use the terms interchangeably. If you say "*estimated* residuals," then it will always be clear that you are referring to $\hat{\varepsilon}_i$.

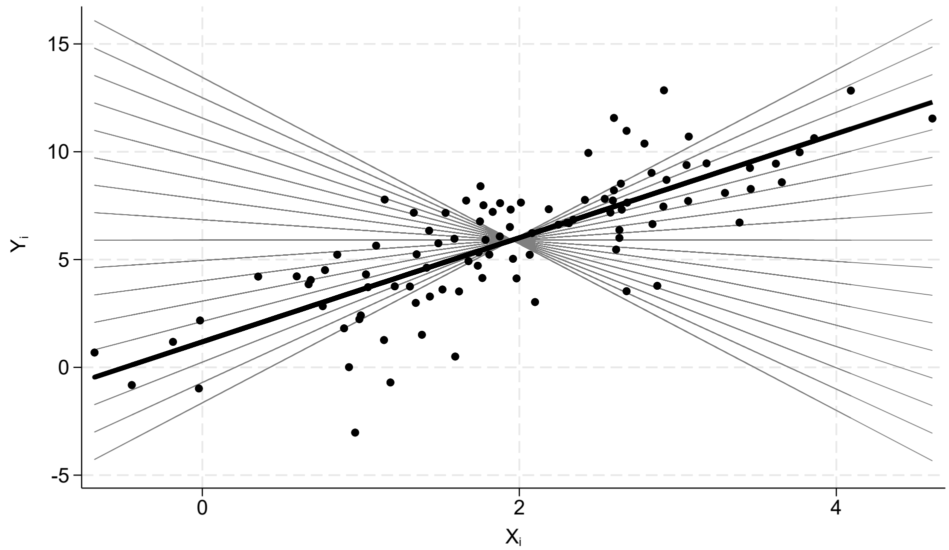

Figure 2: Why Do Orthogonality Conditions Pin Down OLS?

number of lines that satisfy this condition. But once we add the condition that the estimated residual has zero covariance with the regressor, $\widehat{\text{Cov}}\left(\hat{\varepsilon}_i, X_i\right) = 0$, there is only one line, which we draw thicker than the others, that satisfies both conditions.

Least Squares in the Population

We just derived the least-squares *estimator*, and showed what it estimates in a sample. But what *population parameter* is the least-squares estimator estimating?

There is a population analogue to the least-squares estimator. The **best linear predictor** solves the following problem

$$\min_{\alpha,\beta} \mathbb{E}\left[\left(Y_i - \alpha - \beta \cdot X_i\right)^2\right]$$

Whereas the OLS estimator minimizes the sample mean of the squared residuals, the best linear predictor minimizes the expected value (population mean) of the squared errors.

Through essentially the same math as before, we can show that the best linear predictor solves the same orthogonality conditions but with population means, variances, and covariances

$$0 = \mathbb{E}\left[Y_i - \alpha^{BLP} - \beta^{BLP} \cdot X_i\right]$$
$$0 = \text{Cov}\left(Y_i - \alpha^{BLP} - \beta^{BLP} \cdot X_i, X_i\right)$$

and that the estimator also takes a similar form

$$\alpha^{BLP} = \mathbb{E}\left[Y_i\right] - \beta^{BLP} \cdot \mathbb{E}\left[X_i\right]$$
$$\beta^{BLP} = \frac{\text{Cov}\left(X_i, Y_i\right)}{\text{Var}\left(X_i\right)}$$

We will use this fact in later lectures to show that the OLS estimator is a good estimator of the best linear predictor. Because sample means are well-behaved (for example, they are a consistent and asymptotically normal estimator of the population mean), we will be able to show that OLS inherits many of these nice properties as an estimator.

The least-squares estimator is an example of the **plug-in principle**, which is the idea that a good estimator of a population parameter is often the sample analogue of that parameter. For example, the sample median is a good estimator of the population median, the sample variance is a good estimator of the population variance, etc. When the population parameter is a function of means, the plug-in principle is also called the **method of moments**. Note that the term "moments" includes not just the mean, $\mathbb{E}[X_i]$, but also things like variances, covariances, skewness, etc., since these are also functions of expected values (for example, $\text{Var}(X_i) = \mathbb{E}\left[X_i^2\right] - \mathbb{E}[X_i]^2$). The OLS estimator is also an example of a method-of-moments estimator.

Preview of Next Lecture

Exercise 5. Assume that each observation, (X_i, Y_i), is drawn iid. Show that $\widehat{\text{Cov}}(X_i, Y_i) \xrightarrow{p} \text{Cov}(X_i, Y_i)$ and $\widehat{\text{Var}}(X_i) \xrightarrow{p} \text{Var}(X_i)$. *Hint: Use the fact that the sample covariance and variance are functions of sample means. That is,* $\widehat{Cov}(X_i, Y_i) = \overline{XY} - \bar{X} \cdot \bar{Y}$ *and* $\widehat{Var}(X_i) = \overline{X^2} - \bar{X} \cdot \bar{X}$.

Exercise 6. Assume that $\text{Var}(X_i) > 0$. Use this to show that the OLS estimator, $\hat{\beta}$, converges to the best linear predictor, β^{BLP}.

Least Squares with Multiple Regressors

We now move to the case with multiple regressors; this is often called **multivariate regression**. We now write the regression model as:We write this model as follows:

$$\begin{aligned} Y_i &= \alpha + \beta_1 \cdot X_{1,i} + \cdots + \beta_K \cdot X_{K,i} + \varepsilon_i \\ &= \alpha + \left(\sum_{k=1}^{K} \beta_k \cdot X_{k,i}\right) + \varepsilon_i \end{aligned}$$

where K is the number of regressors. The least-squares estimator is now the solution to the following minimization problem:

$$\min_{\alpha,\beta} \frac{1}{N} \sum_{i=1}^{N} \left(Y_i - \alpha - \sum_{k=1}^{K} \beta_k \cdot X_{k,i}\right)^2$$

We can derive first-order conditions in the same way as before, which yields the same orthogonality conditions:

$$\begin{aligned} 0 &= \frac{1}{N} \sum_{i=1}^{N} \hat{\varepsilon}_i \\ 0 &= \widehat{\text{Cov}}(\hat{\varepsilon}_i, X_{k,i}) \text{ for all } k = 1, ..., K \end{aligned}$$

This yields a system of $K + 1$ linear equations, which is just enough to pin down our $K + 1$ coefficients (the K slope coefficients plus the intercept). As long as none of the equations are linearly dependent on each other (we will talk about this more in the next lecture), this provides a unique solution to the least squares problem.

Conveniently, there is an analytic formula that solves the above orthogonality conditions. This course does not require linear algebra as a prerequisite, so for our purposes it is sufficient to know that OLS is a solution to the orthogonality conditions, and that the computer can compute the solution quickly with an analytic formula.